

Amazon RDS Restore from S3 for Database Migration

To Begin with the Lab

Summary of the Lab

In this lab, the **Amazon RDS** console was used to create a new database using the **Restore from S3** option instead of the regular **Create database** flow. The video explained that this option is useful when migrating a database backup stored as a .sql dump file in **Amazon S3**, especially for **MySQL** and **Aurora MySQL** workloads. The process required an S3 source bucket, an **IAM** role so RDS could read the dump file, and the target RDS instance configuration. The lab also clarified that this option is not meant for restoring automated backups or snapshots, and that **AWS DMS** is the better choice for live migration. Finally, the setup was verified by understanding that RDS reads the dump file from S3, creates the database, and restores schema and data in one flow.

Understanding Restore from S3 in Amazon RDS

Restore from S3 exists so you can create a new RDS database and import data from a .sql dump file in one step. It is different from creating an empty database first and then importing data manually, because RDS handles the initialization and restore together. This feature is especially useful when moving a database from on-premises or another environment into **Amazon RDS**. It is not used for automated backups or snapshots, which have their own restore options.

Example:

A team exports a MySQL database from an on-premises server as backup.sql, uploads it to **Amazon S3**, and then uses **Restore from S3** to create a new RDS MySQL instance. RDS reads the dump file, creates the database, and restores the tables and records automatically. That saves time compared with creating an empty instance and importing the backup separately.

Lab Steps

Step 1 – Create a Sample MySQL Database Locally

- Install **MySQL Server** on your local machine.
- NOTE – Make sure to use compatible or same versions to avoid version mismatch error for MySQL
- Open **MySQL Command Line Client** or **MySQL Workbench**.
- Make sure to switch to SQL mode and connect to SQL.

```
MySQL SQL > \connect root@localhost
Creating a session to 'root@localhost'
Please provide the password for 'root@localhost': *****
Save password for 'root@localhost'? [Y]es/[N]o/[e]x (default No): y
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 14 (X protocol)
Server version: 8.0.46 MySQL Community Server - GPL
No default schema selected; type \use <schema> to set one.
MySQL localhost:33060+ ssl SQL >
```

- Create a sample database.

```
CREATE DATABASE studentdb;
```

- Switch to the database.

```
USE studentdb;
```

- Create a sample table.

```
CREATE TABLE students (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    course VARCHAR(100)  
);
```

- Insert sample records.

```
INSERT INTO students (name, course)  
VALUES  
( 'Sahil', 'AWS Data Engineering'),  
( 'Rahul', 'Cloud Computing');
```

- Verify the data.

```
SELECT * FROM students;
```

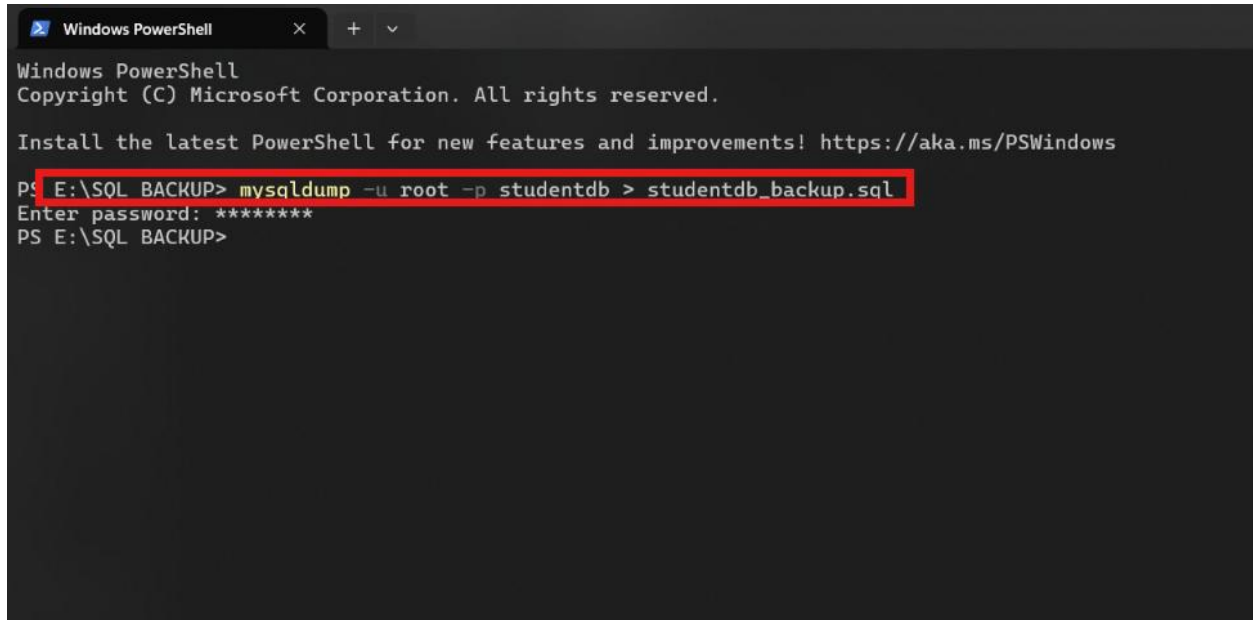
```
MySQL localhost:33060+ ssl SQL > CREATE DATABASE studentdb;  
Query OK, 1 row affected (0.0510 sec)  
MySQL localhost:33060+ ssl SQL > USE studentdb;  
Default schema set to 'studentdb'.  
Fetching global names, object names from 'studentdb' for auto-completion... Press ^C to stop.  
MySQL localhost:33060+ ssl studentdb SQL > CREATE TABLE students (  
-> id INT PRIMARY KEY AUTO_INCREMENT,  
-> name VARCHAR(100),  
-> course VARCHAR(100)  
-> );  
Query OK, 0 rows affected (0.0000 sec)  
MySQL localhost:33060+ ssl studentdb SQL > INSERT INTO students (name, course)  
-> VALUES  
-> ('Sahil', 'AWS Data Engineering'),  
-> ('Rahul', 'Cloud Computing');  
Query OK, 2 rows affected (0.0130 sec)  
Records: 2 Duplicates: 0 Warnings: 0  
MySQL localhost:33060+ ssl studentdb SQL > Verify the data.  
-----  
-----  
SELECT * FROM students at line 1  
MySQL localhost:33060+ ssl studentdb SQL > SELECT * FROM students;  
+----+-----+-----+  
| id | name | course |  
+----+-----+-----+  
| 1 | Sahil | AWS Data Engineering |  
| 2 | Rahul | Cloud Computing |  
+----+-----+-----+  
2 rows in set (0.0439 sec)  
MySQL localhost:33060+ ssl studentdb SQL >
```

Step 2 – Create a MySQL Backup File

- Open Command Prompt or Terminal.
- Navigate to a folder where the backup should be stored.
- Export the database using **mysqldump**.

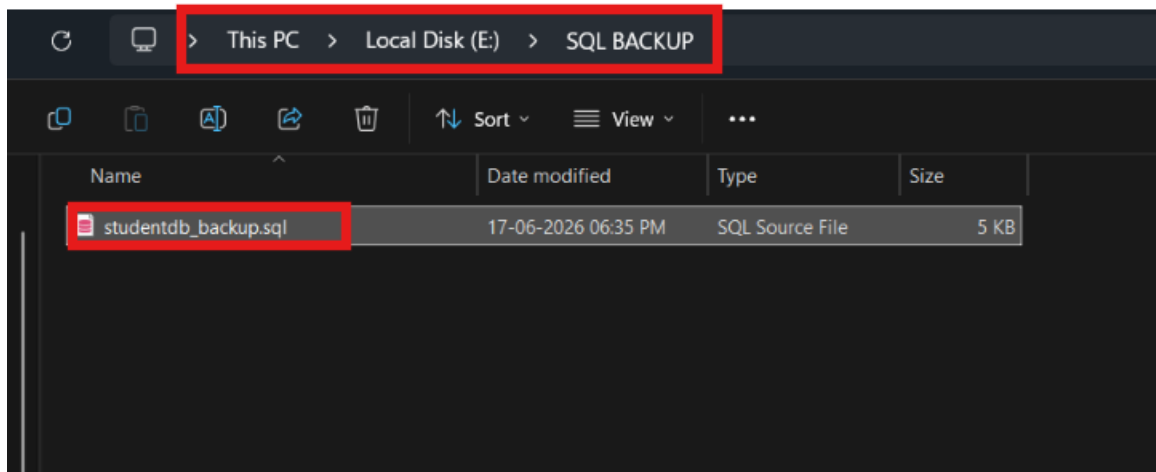
```
mysqldump -u root -p studentdb > studentdb_backup.sql
```

- Enter the MySQL password when prompted.



A screenshot of a Windows PowerShell terminal window. The title bar says "Windows PowerShell". The text inside shows the command prompt at "PS E:\SQL BACKUP>". The command entered is "mysqldump -u root -p studentdb > studentdb_backup.sql", which is highlighted with a red box. Below the command, it says "Enter password: *****" and then "PS E:\SQL BACKUP>".

- Verify that studentdb_backup.sql is created successfully.



Step 3 – Upload the Backup to Amazon S3

- Sign in to the AWS Management Console.
- Open **Amazon S3**.
- Click **Create bucket**.
- Enter a bucket name.

Amazon S3 > Buckets > Create bucket

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

rds-mysql-backup-1

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)** ☐ **ACLs enabled**

- Select the same AWS Region where the RDS instance will be created.
- Click **Create bucket**.
- Open the bucket.
- Click **Upload**.
- Select studentdb_backup.sql.
- Click **Upload**.
- Verify that the SQL file appears in the bucket.

Amazon S3 > Buckets > rds-mysql-backup-1

rds-mysql-backup-1 [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open i2](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	studentdb_backup.sql	sql	June 17, 2026, 18:37:13 (UTC+05:30)	4.1 KB	Standard

Step 4 – Create the IAM Role for RDS

- Open **IAM**.
- Click **Roles**.
- Click **Create role**.
- Select **RDS** as the trusted service.
- Choose **RDS – Add Role to Database**.

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
RDS

Choose a use case for the specified service.

Use case

- ☐ RDS - Directory Service
Allows RDS to manage Directory Service resources on your behalf.
- ☐ RDS - Enhanced Monitoring
Allows RDS to manage CloudWatch Logs resources for Enhanced Monitoring on your behalf.
- ☒ **RDS - Add Role to Database**
Allows you to grant RDS access to additional resources on your behalf.
- ☐ RDS
Allows RDS to perform operations using AWS resources on your behalf.

- Attach permissions allowing access to the S3 bucket.

Add permissions [Info](#)

Permissions policies (1/2280) [Info](#)

Choose one or more policies to attach to your new role.

Search: s3fu Filter by Type: All types 1 match

Policy name	Type	Description
☒ AmazonS3FullAccess	AWS managed	Provides full access to all buckets via the ...

- Name the role.

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
rds-s3-restore
Maximum 64 characters. Use alphanumeric and "+_=@-." characters.

Description
Add a short explanation for this role.
Allows you to grant RDS access to additional resources on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/[()!#\$%^&*{}~`"

Step 1: Select trusted entities

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "",
6       "Effect": "Allow",
7       "Principal": {
8         "Service": [
9           "rds.amazonaws.com"
10        ]
11      }
12    ]
13  }

```

- Create the role.
- Verify that the role can access the S3 bucket containing the SQL dump.

Step 5 – Restore the Database from S3

- Open Amazon RDS.
- Click **Restore from S3**.
- Select **MySQL** or **Aurora MySQL**.
- Provide the S3 bucket name.

Create database by restoring from S3

S3 source

S3 bucket
Choose the Amazon S3 bucket that contains your database backup files.
rds-mysql-backup-1

S3 prefix - optional | Info
Enter the file path prefix for the files stored in your Amazon S3 bucket.

Engine options

Engine type | Info

☐ Aurora (MySQL Compatible)

☒ MySQL

- Provide the SQL dump file path.
- Select the IAM role created earlier.
- Configure the minimum instance.

Restore from S3

IAM role

IAM role
Create or choose an IAM role with a trust policy and a permissions policy that allows Amazon RDS to access your Amazon S3 bucket. [Learn more](#)
rds-s3-restore [Create new role](#)

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | Info

▼ Hide filters

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type
db.t3.micro

Settings

Edition

☒ MySQL Community

Source engine version | Info

- Configure the master username and password.
- Choose engine and version that matches from the AWS options.
- Example – create backup using MySQL 8.0.45 or below to avoid restore error.

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) cloudriteshk (483645775279) cf-shashank1

Aurora and RDS > Databases > Restore from S3

Edition
MySQL Community

Source engine version Info
8.0

Engine version
MySQL 8.0.45

☐ Enable RDS Extended Support Info
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

DB instance identifier Info
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-1
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username Info
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
☒ Self managed
Specify your master password or have RDS create a password that you manage.

☐ Auto-generate password

- Choose minimum storage.

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) cloudriteshk (483645775279) cf-shashank1

Aurora and RDS > Databases > Restore from S3

Storage

Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage Info
20 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

For high-throughput workloads, we recommend provisioning at least 100 GiB of General Purpose (SSD) storage. Lower values might result in higher latencies when the initial I/O credit balance is used up. [Learn more](#)

After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. You can't make further storage modifications for either six (6) hours or until storage optimization has completed on the instance, whichever is longer. [Learn more](#)

► Additional storage configuration

Availability & durability

Multi-AZ deployment Info
☐ Create a standby instance (recommended for production usage)
Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.
☒ Do not create a standby instance

- Configure networking and security groups.

The screenshot shows the 'Restore from S3' configuration page in the AWS Management Console. The page is for an Amazon RDS database instance. The configuration options are as follows:

- Virtual private cloud (VPC):** A dropdown menu showing 'Default VPC (vpc-099792461c7d0a616)'. A red box highlights this selection.
- DB subnet group:** A dropdown menu showing 'default-vpc-099792461c7d0a616'. A red box highlights this selection.
- Public access:** Radio buttons for 'Yes' and 'No'. 'No' is selected.
- VPC security group (firewall):** Two options: 'Choose existing' (selected) and 'Create new'. Under 'Choose existing', a dropdown menu shows 'RDS-SG' and 'default'. Both are highlighted with red boxes.

- Click **Create database**.
- Wait for the database status to become **Available**.

Verification

- A local MySQL database was created successfully.
- A .sql backup file was generated using **mysqldump**.
- The backup file was uploaded to **Amazon S3**.
- An **IAM** role was created to allow RDS access to the S3 bucket.
- The database was restored using **Restore from S3**.
- The restored RDS database contained the same tables and records as the original MySQL database.